

# Tharun Hrithick SR

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## Objective

Fourth-year B.Tech student in Information Science and Engineering with hands-on experience in Python, Generative AI, and prompt engineering. Skilled at working closely with large language models, writing clear domain-specific prompts, and evaluating model outputs with strong attention to detail. Comfortable working independently in remote, asynchronous environments, while remaining adaptable to in-office settings and eager to apply research, reasoning, and analytical skills to help improve the performance of AI systems.

## Skills

- **Programming:** Python, Java, JavaScript
- **Web Development:** HTML, CSS
- **Database:** MySQL, PostgreSQL, ChromaDB
- **Tools:** Git, n8n, Claude Code, Postman
- **Technical Interests:** Artificial Intelligence, Agentic AI, Prompt Engineering, Machine Learning, Generative AI, RAG, Langchain, LangGraph
- **Soft Skills:** Problem-solving, Collaboration, Adaptability, Quick Learning

## Projects

### Early Lung Cancer Detection Web App -

(Completed)

Deep Learning | Flask | PyTorch | CT Scans | SQL

- Designed an end-to-end medical image classification system using Convolutional Neural Networks (CNNs) to detect lung cancer from CT scan images.
- Integrated the trained CNN model with a Flask backend and SQL database, achieving 91% validation accuracy while enabling real-time diagnostic predictions.

### Nerve – AI Healthcare Assistant -

(Completed)

FastAPI | React | TypeScript | PostgreSQL | ChromaDB | Mistral AI | RAG | OCR | Docker

- Built a full-stack AI healthcare assistant using Retrieval-Augmented Generation (RAG) to answer medical queries from uploaded reports, PDFs, and images with context-aware responses.
- Implemented OCR-based document processing, semantic search with ChromaDB, JWT/Google authentication, and persistent chat history using FastAPI and PostgreSQL.

### Automated Report Generator -

(Completed)

FastAPI | Generative AI | Mistral API | SQLAlchemy | SQLite

- Built an intelligent automated reporting tool that parses active Git repositories to auto-generate structured development summaries via the Mistral API.

### Emotion Detection & Music Recommendation -

(Completed)

PyTorch | ONNX Runtime | Scikit-Learn | Flask | Python | Cosine Similarity | Tailwind CSS

- Built a custom **Bidirectional LSTM (BiLSTM)** classification model in **PyTorch** to detect 6 primary emotional states from raw text inputs, utilizing L2 regularization, gradient clipping, and learning rate scheduling to ensure stable convergence.

## Education

### B.Tech in Information Science and Engineering

Jain (Deemed-to-be University), Bangalore

(2023 - 2027)

- Currently in 4th Year | Aggregate CGPA: 7.55

### Paavai Vidhyashram School, Namakkal, Tamil Nadu 637014

Higher Secondary Education (12th) PCMB CBSE, 2023 - 66% | Secondary School Education (10th), 2021 - 69%

## Internships -

### Agentic AI Development Intern — MacAppStudio

(09.07.2026 - 30.07.2026)

- Developed Nerve, an AI-powered Healthcare Assistant using FastAPI, React, and Retrieval-Augmented Generation (RAG) to provide context-aware responses from uploaded medical reports, PDFs, and images.
- Implemented OCR-based document processing with Tesseract, semantic retrieval using ChromaDB, JWT/Google authentication, and PostgreSQL-backed conversation management for secure, persistent user interactions.

### Artificial Intelligence virtual intern — Infosys Springboard 6.0

(15.09.2025 - 25.11.2025)

- Engineered a dual-module pipeline combining computer vision (**OpenCV / CNNs**) for facial expression detection and NLP classification (**TF-IDF / SVM**) for text sentiment analysis, achieving an aggregate classification accuracy of **88%**.
- Structured data preprocessing pipelines for metadata normalization, improving overall music recommendation relevance metrics.

### Backend Development (Python) — Hawky (Startup), Bangalore

(09.06.2025 - 30.07.2025)

- Architected and optimized high-throughput Flask API endpoints, reducing core application response latencies by 15%.
- Collaborated with startup engineers to build scalable database schemas and resolve critical server bottlenecks prior to product launch.

## Certifications -

- Generative AI — Microsoft & LinkedIn
- ChatGPT Prompt Engineering — DeepLearning.AI
- Artificial Intelligence — Infosys Springboard
- Introduction to Agent Skills — Anthropic
- Database and SQL for Data Science in Python — Coursera (IBM)